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$$\begin{aligned}& \int \frac{x}{x^2 + 2} dx \\& u = x^2 + 2 \Rightarrow \frac{du}{dx} = 2x \Rightarrow du = 2x dx \Rightarrow x dx = \frac{1}{2} du \\& \Rightarrow \int \frac{x}{x^2 + 2} dx = \int \frac{1}{u} \cdot \frac{1}{2} du \Rightarrow \frac{1}{2} \int \frac{1}{u} du \\& \Rightarrow \int \frac{1}{u} du = \ln |u| \Rightarrow \frac{1}{2} \ln |u| + C \\& \Rightarrow u = x^2 + 2 \Rightarrow \frac{1}{2} \ln(x^2 + 2) + C \\& \Rightarrow \int \frac{x}{x^2 + 2} dx = \frac{1}{2} \ln(x^2 + 2) + C\end{aligned}$$

References